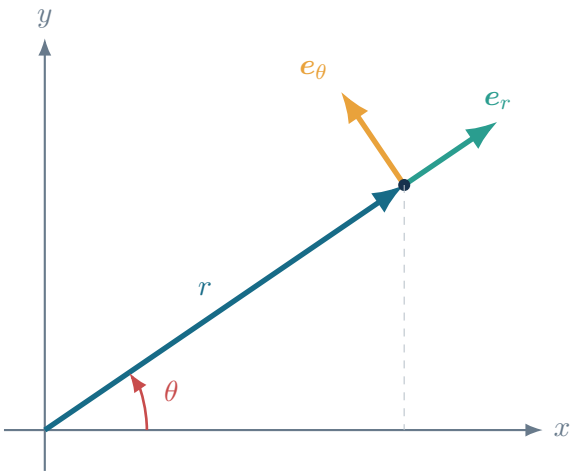
Vectors in Polar Coordinates

A very slow mechanics tutorial

From tiny displacements to circular orbits and the law $P \propto r^{3/2}$



The position changes in two independent ways: outward or inward and around the origin.

The single idea to keep in mind

The Cartesian directions $\boldsymbol{i}$ and $\boldsymbol{j}$ stay fixed. The polar directions $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ rotate whenever θ changes. Most of the apparently new terms in polar acceleration come from differentiating these rotating directions carefully.

Notation note: K will mean kinetic energy and P will mean orbital period, so the same letter is never asked to do two jobs.

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1 What polar coordinates describe

In a plane, a point can be located by either Cartesian coordinates (x, y) or polar coordinates (r, θ). The two descriptions refer to the same point:

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Here $r \geq 0$ is the distance from the origin and θ is an angle measured counterclockwise from the positive x-axis. In mechanics, both may depend on time:

$$r = r(t), \qquad \theta = \theta(t).$$

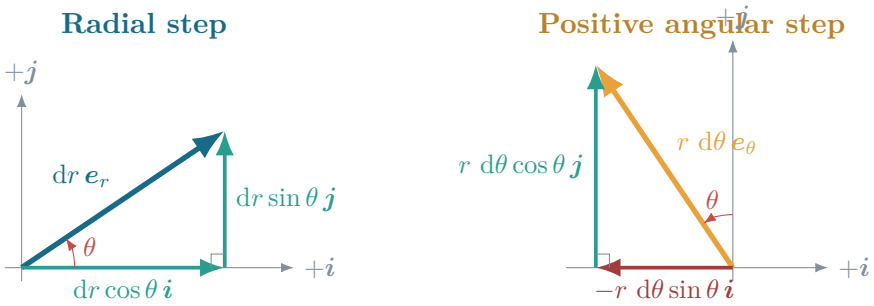
1.1 The two polar unit vectors

At the particle we attach two perpendicular unit vectors:

- $\boldsymbol{e}_r$ points directly away from the origin;
- $\boldsymbol{e}_\theta$ points in the direction of increasing θ, tangent to the circle $r =$ constant.

1.2 Deriving the unit vectors from little triangles

The component formulas do not need to be memorised. Draw a short displacement, complete a right triangle, and read off its horizontal and vertical sides.



The radial triangle. Take a tiny outward step of length $\mathrm{d}r$ at angle θ. Its horizontal side is $\mathrm{d}r\cos\theta$ and its vertical side is $\mathrm{d}r\sin\theta$. Hence

$$\begin{aligned}\mathrm{d}\boldsymbol{r}_{\text{radial}} &= \mathrm{d}r\cos\theta\,\boldsymbol{i} + \mathrm{d}r\sin\theta\,\boldsymbol{j}\\ &= \mathrm{d}r\,(\cos\theta\,\boldsymbol{i} + \sin\theta\,\boldsymbol{j})\,.\end{aligned}$$

Dividing by the step length $\mathrm{d}r$ leaves the radial unit direction:

$$\boldsymbol{e}_r = \cos\theta\,\boldsymbol{i} + \sin\theta\,\boldsymbol{j}.$$

The tangential triangle. Now increase the angle while keeping r fixed. The tiny arc has length $r\,\mathrm{d}\theta$ and points 90° counterclockwise from $\boldsymbol{e}_r$. In the first quadrant it points left and up: its horizontal component is negative, with magnitude $r\,\mathrm{d}\theta\sin\theta$, while its vertical component is positive, with magnitude $r\,\mathrm{d}\theta\cos\theta$. Therefore

$$\begin{aligned}\mathrm{d}\boldsymbol{r}_{\text{angular}} &= -r\,\mathrm{d}\theta\sin\theta\,\boldsymbol{i} + r\,\mathrm{d}\theta\cos\theta\,\boldsymbol{j}\\ &= r\,\mathrm{d}\theta\,(-\sin\theta\,\boldsymbol{i} + \cos\theta\,\boldsymbol{j})\,.\end{aligned}$$

Dividing by the tangential step length $r\,\mathrm{d}\theta$ gives the positive angular unit direction:

$$\boldsymbol{e}_\theta = -\sin\theta\,\boldsymbol{i} + \cos\theta\,\boldsymbol{j}.$$

Where the minus sign comes from

For $0 < \theta < \pi/2$, the positive tangential direction goes left and up. Left means a negative $\boldsymbol{i}$ component; up means a positive $\boldsymbol{j}$ component. The same formula continues to work in every quadrant.

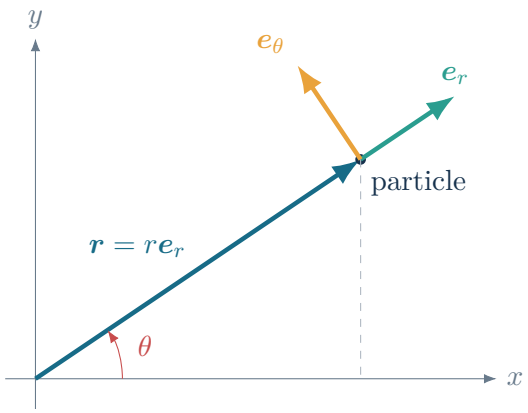
Collecting the two results,

Polar basis written in the Cartesian basis

$$\boxed{\boldsymbol{e}_r = \cos\theta\,\boldsymbol{i} + \sin\theta\,\boldsymbol{j}}, \qquad \boxed{\boldsymbol{e}_\theta = -\sin\theta\,\boldsymbol{i} + \cos\theta\,\boldsymbol{j}}.$$

These are unit vectors and they are perpendicular:

$$\boldsymbol{e}_r \cdot \boldsymbol{e}_r = 1, \qquad \boldsymbol{e}_\theta \cdot \boldsymbol{e}_\theta = 1, \qquad \boldsymbol{e}_r \cdot \boldsymbol{e}_\theta = 0.$$



A useful warning

The symbols $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ do *not* name two globally fixed directions. Their orientation depends on θ. If the particle moves around the origin, the basis itself turns.

2 Tiny displacements: one radial, one angular

Suppose the particle starts at (r, θ). We first change one coordinate at a time. This separates the geometry before we combine it.

2.1 Step 1: change r, but hold θ fixed

Move from (r, θ) to $(r + \mathrm{d}r, \theta)$. The motion is along the radial line, so

$$\mathrm{d}\boldsymbol{r}_{\text{radial}} = \mathrm{d}r\, \boldsymbol{e}_r.$$

The scalar $\mathrm{d}r$ is a signed length. If $\mathrm{d}r > 0$, the displacement is outward; if $\mathrm{d}r < 0$, it is inward.

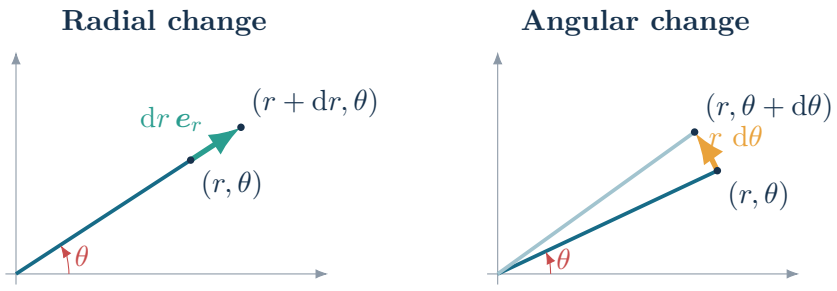
2.2 Step 2: change θ, but hold r fixed

Now move from (r, θ) to $(r, \theta + \mathrm{d}\theta)$. The particle follows a tiny circular arc. Its length is

$$\text{arc length} = r\ \mathrm{d}\theta,$$

provided $\mathrm{d}\theta$ is measured in radians. The tangent direction is $\boldsymbol{e}_\theta$, so

$$\mathrm{d}\boldsymbol{r}_{\text{angular}} = r\ \mathrm{d}\theta\, \boldsymbol{e}_\theta.$$



2.3 Step 3: add the two independent changes

To first order, a general tiny displacement is the sum:

Differential displacement in polar coordinates

$$\boxed{\mathrm{d}\boldsymbol{r} = \mathrm{d}r\, \boldsymbol{e}_r + r\ \mathrm{d}\theta\, \boldsymbol{e}_\theta.}$$

The two coefficients have units of length: $\mathrm{d}r$ is a length, while $r\,\mathrm{d}\theta$ is also a length because an angle in radians is dimensionless.

Why a tiny change matters

For a finite angular change, the straight chord and the circular arc are not identical. Their difference enters beyond first order. Differential calculus keeps the terms linear in $\mathrm{d}r$ and $\mathrm{d}\theta$, which is exactly what is needed for velocity.

3 Velocity

Divide the differential displacement by the elapsed time $\mathrm{d}t$:

$$\begin{aligned}\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} &= \frac{\mathrm{d}r}{\mathrm{d}t}\,\boldsymbol{e}_r + r\frac{\mathrm{d}\theta}{\mathrm{d}t}\,\boldsymbol{e}_\theta \\ &= \dot{r}\,\boldsymbol{e}_r + r\dot{\theta}\,\boldsymbol{e}_\theta.\end{aligned}$$

Polar velocity

$$\boxed{\boldsymbol{v} = \dot{r}\,\boldsymbol{e}_r + r\dot{\theta}\,\boldsymbol{e}_\theta.}$$

The two pieces have immediate meanings:

- $\dot{r}\,\boldsymbol{e}_r$ is the radial velocity;
- $r\dot{\theta}\,\boldsymbol{e}_\theta$ is the tangential velocity.

3.1 A Cartesian cross-check

Start from

$$\boldsymbol{r} = x\boldsymbol{i} + y\boldsymbol{j}, \qquad x = r\cos\theta, \qquad y = r\sin\theta.$$

Differentiate the scalar coordinates:

$$\begin{aligned}\dot{x} &= \dot{r}\cos\theta - r\dot{\theta}\sin\theta, \\ \dot{y} &= \dot{r}\sin\theta + r\dot{\theta}\cos\theta.\end{aligned}$$

Therefore

$$\begin{aligned}\boldsymbol{v} &= \dot{x}\boldsymbol{i} + \dot{y}\boldsymbol{j} \\ &= \dot{r}(\cos\theta\,\boldsymbol{i} + \sin\theta\,\boldsymbol{j}) + r\dot{\theta}(-\sin\theta\,\boldsymbol{i} + \cos\theta\,\boldsymbol{j}) \\ &= \dot{r}\,\boldsymbol{e}_r + r\dot{\theta}\,\boldsymbol{e}_\theta.\end{aligned}$$

The Cartesian calculation agrees with the geometric one. This is a useful check, but the polar form makes the physical directions much easier to see.

Pause and picture it

If $\dot{r} = 0$, the particle has no inward or outward motion, but it can still be moving: $r\dot{\theta}$ is its speed around the circle. If $\dot{\theta} = 0$, the direction from the origin is fixed, but the particle may still move radially.

4 The basis vectors rotate

It is tempting to differentiate the velocity as if $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ were constants. They are not. We now calculate their derivatives explicitly.

4.1 Differentiate with respect to the angle

Using their Cartesian expressions,

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}\theta} &= \frac{\mathrm{d}}{\mathrm{d}\theta}(\cos\theta\,\boldsymbol{i} + \sin\theta\,\boldsymbol{j}) \\ &= -\sin\theta\,\boldsymbol{i} + \cos\theta\,\boldsymbol{j} \\ &= \boldsymbol{e}_\theta,\end{aligned}$$

and

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{e}_\theta}{\mathrm{d}\theta} &= \frac{\mathrm{d}}{\mathrm{d}\theta}(-\sin\theta\,\boldsymbol{i} + \cos\theta\,\boldsymbol{j}) \\ &= -\cos\theta\,\boldsymbol{i} - \sin\theta\,\boldsymbol{j} \\ &= -\boldsymbol{e}_r.\end{aligned}$$

How the polar basis turns

$$\boxed{\frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}\theta} = \boldsymbol{e}_\theta}, \qquad \boxed{\frac{\mathrm{d}\boldsymbol{e}_\theta}{\mathrm{d}\theta} = -\boldsymbol{e}_r}.$$

The first formula says that a small increase in angle tips $\boldsymbol{e}_r$ toward $\boldsymbol{e}_\theta$. The second says that the tangent direction bends inward, toward $-\boldsymbol{e}_r$.

4.2 Convert angle derivatives to time derivatives

Apply the chain rule:

$$\frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t} = \dot\theta\,\boldsymbol{e}_\theta,$$

and similarly

$$\frac{\mathrm{d}\boldsymbol{e}_\theta}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{e}_\theta}{\mathrm{d}\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\dot\theta\,\boldsymbol{e}_r.$$

Time derivatives of the rotating basis

$$\boxed{\dot{\boldsymbol{e}}_r = \dot\theta\,\boldsymbol{e}_\theta}, \qquad \boxed{\dot{\boldsymbol{e}}_\theta = -\dot\theta\,\boldsymbol{e}_r}.$$

5 Acceleration, one product rule at a time

Begin with the velocity:

$$\boldsymbol{v} = \dot{r}\,\boldsymbol{e}_r + r\dot{\theta}\,\boldsymbol{e}_\theta.$$

Differentiate each vector term separately.

5.1 Step 1: differentiate the radial velocity term

Use the product rule and $\dot{\boldsymbol{e}}_r = \dot{\theta}\boldsymbol{e}_\theta$:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(\dot{r}\,\boldsymbol{e}_r) &= \ddot{r}\,\boldsymbol{e}_r + \dot{r}\,\dot{\boldsymbol{e}}_r \\ &= \ddot{r}\,\boldsymbol{e}_r + \dot{r}\dot{\theta}\,\boldsymbol{e}_\theta.\end{aligned}$$

5.2 Step 2: differentiate the tangential velocity term

The scalar coefficient $r\dot{\theta}$ and the vector $\boldsymbol{e}_\theta$ both vary:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(r\dot{\theta}\,\boldsymbol{e}_\theta) &= \frac{\mathrm{d}}{\mathrm{d}t}(r\dot{\theta})\,\boldsymbol{e}_\theta + r\dot{\theta}\,\dot{\boldsymbol{e}}_\theta \\ &= (\dot{r}\dot{\theta} + r\ddot{\theta})\boldsymbol{e}_\theta + r\dot{\theta}(-\dot{\theta}\boldsymbol{e}_r) \\ &= (\dot{r}\dot{\theta} + r\ddot{\theta})\boldsymbol{e}_\theta - r\dot{\theta}^{\,2}\boldsymbol{e}_r.\end{aligned}$$

5.3 Step 3: collect like directions

Add the two results. The $\boldsymbol{e}_r$ terms go together and the $\boldsymbol{e}_\theta$ terms go together:

$$\begin{aligned}\boldsymbol{a} &= \left(\ddot{r} - r\dot{\theta}^{\,2}\right)\boldsymbol{e}_r + \left(\dot{r}\dot{\theta} + \dot{r}\dot{\theta} + r\ddot{\theta}\right)\boldsymbol{e}_\theta \\ &= \left(\ddot{r} - r\dot{\theta}^{\,2}\right)\boldsymbol{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\boldsymbol{e}_\theta.\end{aligned}$$

Polar acceleration

$$\boxed{\boldsymbol{a} = \left(\ddot{r} - r\dot{\theta}^{\,2}\right)\boldsymbol{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\boldsymbol{e}_\theta.}$$

The common mistake

Writing only $\ddot{r}\boldsymbol{e}_r + r\ddot{\theta}\boldsymbol{e}_\theta$ misses the derivatives of the basis vectors. The missing terms, $-r\dot{\theta}^2\boldsymbol{e}_r$ and $2\dot{r}\dot{\theta}\boldsymbol{e}_\theta$, are not optional corrections; they are required by the geometry.

6 What the four acceleration terms mean

The formula is easier to remember when each term has a job:

Term	Direction	Interpretation
$\ddot{r}$	radial	Change in the radial speed. Positive is outward.
$-r\dot{\theta}^2$	radial	Inward curvature term. It remains even when the speed around a circle is constant.
$r\ddot{\theta}$	transverse	Tangential acceleration caused by changing angular speed.
$2\dot{r}\dot{\theta}$	transverse	Coupling between radial motion and rotation of the basis, often called the Coriolis term in this coordinate description.

6.1 A quick dimensional check

Angles in radians are dimensionless. Therefore

$$[\dot{r}] = LT^{-1}, \qquad [\ddot{r}] = LT^{-2}, \qquad [\dot{\theta}] = T^{-1}, \qquad [\ddot{\theta}] = T^{-2}.$$

Every coefficient in the acceleration formula has units LT^{-2}:

$$[r\dot{\theta}^2] = L(T^{-1})^2 = LT^{-2},$$

$$[\dot{r}\dot{\theta}] = (LT^{-1})(T^{-1}) = LT^{-2}, \qquad [r\ddot{\theta}] = L(T^{-2}) = LT^{-2}.$$

Why the inward term appears at constant speed

A velocity vector can change even if its magnitude does not. During circular motion the velocity continually turns, so its derivative points toward the centre. The term $-r\dot{\theta}^2\boldsymbol{e}_r$ measures exactly that inward turning.

6.2 Three useful special cases

1. **Pure radial motion:** if $\dot{\theta} = 0$, then $\boldsymbol{a} = \ddot{r}\boldsymbol{e}_r$.
2. **Motion at fixed radius:** if $r = R$, then
$$\boldsymbol{a} = -R\dot{\theta}^2\boldsymbol{e}_r + R\ddot{\theta}\boldsymbol{e}_\theta.$$
3. **Uniform circular motion:** if $r = R$ and $\dot{\theta} = \omega$ is constant, then
$$\boldsymbol{a} = -R\omega^2\boldsymbol{e}_r.$$

7 Speed squared and kinetic energy

Start again from

$$\boldsymbol{v} = \dot{r}\boldsymbol{e}_r + r\dot{\theta}\boldsymbol{e}_\theta.$$

The square of the speed means the dot product $v^2 = \boldsymbol{v} \cdot \boldsymbol{v}$:

$$\begin{aligned} v^2 &= (\dot{r}\boldsymbol{e}_r + r\dot{\theta}\boldsymbol{e}_\theta) \cdot (\dot{r}\boldsymbol{e}_r + r\dot{\theta}\boldsymbol{e}_\theta) \\ &= \dot{r}^2(\boldsymbol{e}_r \cdot \boldsymbol{e}_r) + 2r\dot{r}\dot{\theta}(\boldsymbol{e}_r \cdot \boldsymbol{e}_\theta) + r^2\dot{\theta}^2(\boldsymbol{e}_\theta \cdot \boldsymbol{e}_\theta). \end{aligned}$$

Now use orthonormality:

$$\boldsymbol{e}_r \cdot \boldsymbol{e}_r = 1, \qquad \boldsymbol{e}_r \cdot \boldsymbol{e}_\theta = 0, \qquad \boldsymbol{e}_\theta \cdot \boldsymbol{e}_\theta = 1.$$

The cross term vanishes, leaving

Speed in polar coordinates

$$\boxed{v^2 = \dot{r}^2 + r^2\dot{\theta}^2.}$$

For a particle of mass m, the kinetic energy is

Kinetic energy in polar coordinates

$$\boxed{K = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right).}$$

For circular motion $r = R$ is constant, so $\dot{r} = 0$. If $\dot{\theta} = \omega$, then

$$v = R\omega, \qquad K = \frac{1}{2}mR^2\omega^2.$$

Dimensions of the energy

Both $\dot{r}^2$ and $r^2\dot{\theta}^2$ have units L^2T^{-2}. Multiplication by mass gives ML^2T^{-2}, the dimensions of energy.

8 Uniform circular motion and centripetal force

The phrase *uniform circular motion* contains several assumptions. We state them before using the shortened formulas.

- The radius is constant: $r(t) = R$. Hence $\dot{r} = 0$ and $\ddot{r} = 0$.
- The angular speed is constant: $\dot{\theta} = \omega$. Hence $\ddot{\theta} = 0$.
- The only acceleration is radial and inward.

Substitute these conditions into the general velocity and acceleration:

$$\boldsymbol{v} = R\omega \boldsymbol{e}_\theta, \qquad \boldsymbol{a} = -R\omega^2 \boldsymbol{e}_r.$$

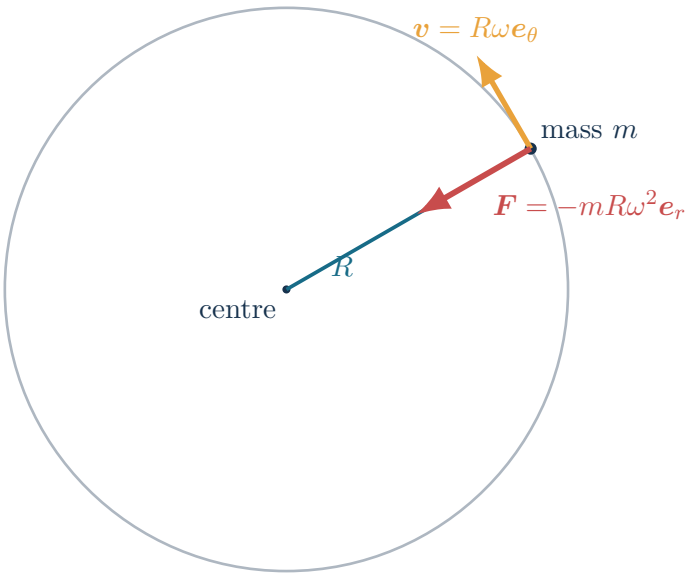
Newton's second law gives

Force required for uniform circular motion

$$\boxed{\boldsymbol{F} = m\boldsymbol{a} = -mR\omega^2 \boldsymbol{e}_r.}$$

The minus sign says that the force points opposite to $\boldsymbol{e}_r$, toward the centre. Its magnitude is

$$F_{\text{in}} = mR\omega^2 = \frac{mv^2}{R}.$$



Circular motion is a special solution

A central force does not automatically make every orbit circular. A circular orbit requires the correct initial tangential speed and no initial radial speed. Other initial conditions can produce non-circular bound or unbound orbits.

9 Inverse-square attraction and the orbital period

We now apply the circular-motion result to an attractive inverse-square field. It is safest to distinguish an *acceleration law* from a *force law*.

9.1 Form A: inverse-square acceleration

For gravity around a fixed central mass M, define

$$\mu = GM.$$

The gravitational acceleration is

$$\boldsymbol{a}_{\text{grav}} = -\frac{\mu}{r^2}\boldsymbol{e}_r.$$

For a circular orbit of radius R, the required acceleration is $-R\omega^2\boldsymbol{e}_r$. Equate their magnitudes:

$$\begin{aligned}\frac{\mu}{R^2} &= R\omega^2,\\ \omega^2 &= \frac{\mu}{R^3},\\ \omega &= \sqrt{\frac{\mu}{R^3}}.\end{aligned}$$

One revolution is 2π radians, so the period is

$$P = \frac{2\pi}{\omega}.$$

Substitution gives

Period of a circular inverse-square orbit

$$\boxed{P = 2\pi\sqrt{\frac{R^3}{\mu}}}, \qquad \boxed{P^2 = \frac{4\pi^2}{\mu}R^3}.$$

Therefore, with the same central body fixed,

$$\boxed{P \propto R^{3/2}.}$$

9.2 Dimensional check

The gravitational parameter has dimensions

$$[\mu] = [GM] = L^3T^{-2}.$$

Therefore

$$\left[\frac{R^3}{\mu}\right] = \frac{L^3}{L^3T^{-2}} = T^2,$$

and its square root has units of time, as a period must.

9.3 Form B: inverse-square force

Suppose instead that the force itself is written

$$\boldsymbol{F} = -\frac{\kappa}{r^2}\boldsymbol{e}_r.$$

Then κ must have dimensions ML^3T^{-2}. On a circular orbit,

$$\begin{aligned}\frac{\kappa}{R^2} &= mR\omega^2,\\ \omega^2 &= \frac{\kappa}{mR^3},\\ P &= 2\pi\sqrt{\frac{mR^3}{\kappa}}.\end{aligned}$$

For gravity, $\kappa = GMm = \mu m$, so the orbiting mass cancels:

$$P = 2\pi\sqrt{\frac{mR^3}{GMm}} = 2\pi\sqrt{\frac{R^3}{GM}}.$$

How this matches shorthand notation

If a handwritten derivation writes $k/R^2 = R\omega^2$, then k is being used as an acceleration parameter with dimensions L^3T^{-2}, like $\mu = GM$. If it writes a force k/R^2, an additional factor of m belongs in Newton's second law unless m has already been absorbed into k.

9.4 What has, and has not, been shown

The calculation above establishes the $R^{3/2}$ scaling for circular orbits in an inverse-square field. Kepler's third law also holds for elliptical two-body orbits when R is replaced by the semi-major axis a:

$$P^2 = \frac{4\pi^2}{\mu}a^3$$

in the fixed-central-mass approximation. For the full two-body problem, $\mu = G(M + m)$. Those broader results require additional orbital mechanics; the circular derivation here captures the scaling in its simplest setting.

10 The entire derivation on one page

1. Define the rotating polar basis:

$$\boldsymbol{e}_r = \cos\theta\boldsymbol{i} + \sin\theta\boldsymbol{j}, \qquad \boldsymbol{e}_\theta = -\sin\theta\boldsymbol{i} + \cos\theta\boldsymbol{j}.$$

2. Resolve a tiny displacement into radial and angular pieces:

$$\mathrm{d}\boldsymbol{r} = \mathrm{d}r\boldsymbol{e}_r + r\,\mathrm{d}\theta\boldsymbol{e}_\theta.$$

3. Divide by time:

$$\boldsymbol{v} = \dot{r}\boldsymbol{e}_r + r\dot{\theta}\boldsymbol{e}_\theta.$$

4. Account for the rotating basis:

$$\dot{\boldsymbol{e}}_r = \dot{\theta}\boldsymbol{e}_\theta, \qquad \dot{\boldsymbol{e}}_\theta = -\dot{\theta}\boldsymbol{e}_r.$$

5. Differentiate the velocity:

$$\boldsymbol{a} = (\ddot{r} - r\dot{\theta}^2)\boldsymbol{e}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\boldsymbol{e}_\theta.$$

6. Use orthonormality to find speed and kinetic energy:

$$v^2 = \dot{r}^2 + r^2\dot{\theta}^2, \qquad K = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2).$$

7. For uniform circular motion, set $r = R$ and $\dot{\theta} = \omega$ constant:

$$\boldsymbol{a} = -R\omega^2\boldsymbol{e}_r, \qquad \boldsymbol{F} = -mR\omega^2\boldsymbol{e}_r.$$

8. Match this to inverse-square gravitational acceleration:

$$\frac{\mu}{R^2} = R\omega^2 \quad\Longrightarrow\quad P = 2\pi\sqrt{\frac{R^3}{\mu}} \quad\Longrightarrow\quad P \propto R^{3/2}.$$

The conceptual thread

The polar formulas are not mysterious new laws. They are ordinary vector differentiation expressed in a basis that turns with θ. Once the basis derivatives are included, velocity, acceleration, energy, and circular-orbit dynamics follow in a continuous chain.

End of tutorial